

Which Equation Fits Best?

Name: _____

Date: _____

The data

A small dataframe has four features, x_1, x_2, x_3, x_4 , and a target y . We want an equation of the form

$$\hat{y} = b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4$$

that predicts y as accurately as possible.

row	x_1	x_2	x_3	x_4	y
0	1	2	3	1	9
1	2	1	3	0	7
2	3	3	6	2	17
3	4	1	5	0	11
4	5	2	7	1	17

Three people each fit a model to this data and report an equation:

- Equation A: $\hat{y} = 2x_1 + 3x_2 + 0x_3 + 1x_4$
- Equation B: $\hat{y} = 1x_1 + 2x_2 + 1x_3 + 1x_4$
- Equation C: $\hat{y} = 0x_1 + 1x_2 + 2x_3 + 1x_4$

Part 1: Check the fits

1. Use each equation to predict y for every row. Fill in the table, then compute the sum of squared errors for each equation, $\text{SSE} = \sum_i (y_i - \hat{y}_i)^2$, summing over all five rows.

row	actual y	A prediction	B prediction	C prediction
0	9			
1	7			
2	17			
3	11			
4	17			
SSE				

2. Which equation is the best fit?

Part 2: Make sense of it

3. Explain what you found? Use the numbers in the data table to support your answer.
4. Compare the coefficient on x_4 across the three equations with the coefficients on x_1 , x_2 , and x_3 . What do you notice? What is different about x_4 ?
5. Is there a fourth equation that also fits this data perfectly? Write one down, and check it against at least two rows.
6. What does this example imply about interpreting the relationship between features and the target variable in a regression problem?

Optional: check it in code

```
import numpy as np
import pandas as pd

df = pd.DataFrame({
    "x1": [1, 2, 3, 4, 5],
    "x2": [2, 1, 3, 1, 2],
    "x3": [3, 3, 6, 5, 7],
    "x4": [1, 0, 2, 0, 1],
    "y": [9, 7, 17, 11, 17],
})

X = df[["x1", "x2", "x3", "x4"]].to_numpy()
print(np.linalg.matrix_rank(X))      # how many independent columns?
print(np.linalg.cond(X))             # condition number
```